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F. An Ancient Sorting Algorithm

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: `sort`, `open`, `file`, `creat`(, `fstream`, `thread`, `process`, `system`(, `exec`(

You are given an array of N integers. You have to sort the array in non-decreasing order using a custom sorting algorithm with the following constraint:

- You may only swap adjacent elements with the same parity (i.e., both even or both odd)

Sort the array in non-decreasing order until no more such swaps are possible and print the final array.

Input

The first line contains a single integer N ($1 \leq N \leq 1000$) — the number of elements in the array.

The second line contains N integers separated by spaces $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq 10^6$) — the elements of the array.

Output

Print the final array after sorting the array in non-decreasing order until no more such swaps are possible.

Examples

input	Copy
7 4 2 4 7 1 6 1	
output	Copy
2 4 4 1 7 6 1	
input	Copy
5 3 5 9 7 1	
output	Copy
1 3 5 7 9	
input	Copy
14 4 8 2 9 1 5 4 6 8 1 7 13 11 8	
output	Copy
2 4 8 1 5 9 4 6 8 1 7 11 13 8	
input	Copy
1 221	
output	Copy

CSE221 Assignment 01 Summer 2026

Contest is running

2 days

Contestant



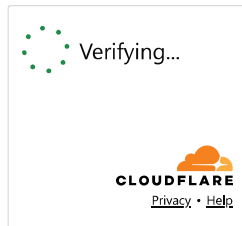
→ Submit?

Language: Java 21 64bit

Choose file:

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No file chosen

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